Krein-pointed proper quantum metric spaces and the bridge to Mondino–Sämann Lorentzian pre-length spaces, with application to the GeoVac hemispheric wedge

J. Loutey*

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Abstract

We construct a categorical bridge between Latrémolière's pinned proper quantum metric space (PPQMS) framework [25] and Mondino–Sämann's covered Lorentzian pre-length space (LPLS) framework [30]. The bridge is a contravariant functor

$$\mathbf{W} : \mathbf{KreinMetaMet}_{\mathrm{pp}} \longrightarrow \mathbf{LorPLG}_{\mathrm{cov}}$$

sending a Krein-pointed proper QMS $(\mathcal{A}^K, L^K, \mathcal{M}^L, \omega^L_W)$ to a covered LPLS $(\hat{\mathcal{M}}^L, \ell^L, \hat{\omega}^L_W, \hat{\mathcal{U}}^L)$ via Gelfand spectrum of the M-diagonal Abelian topography, Wick-rotated modular flow as time separation, the Bisognano–Wichmann vacuum as basepoint, and admissible-scaling truncated covers. The structural mismatch between Latrémolière's 4-point relaxed forward metametric and Mondino–Sämann's 3-point reverse triangle (the F2 mismatch) is resolved via the Connes–Rovelli thermal-time hypothesis [9]: the Krein metametric measures *thermal time* of the wedge KMS state, the synthetic time separation measures *geometric proper time*, and the Wick rotation between them effects the analytic continuation that maps the additive (forward-triangle) thermal-time structure to the super-additive (reverse-triangle) geometric-time structure.

Main theorem. The Wick-rotation functor $\mathbf{W}$ is well-defined and the Krein–Mondino–Sämann Bridge Theorem (Theorem 5.4) holds at theorem-grade rigor: (B1) structural correspondence on 15 rows, (B2) reverse triangle inequality via on-orbit modular-flow equality plus an M-diagonal topography trivialization off-orbit, (B3) pre-compactness inheritance via Paper 44 [48] propagation-number bound, (B4) convergence transport via Plancherel-nested Berezin compatibility with the Mondino–Sämann Def 3.6 axioms.

GeoVac wedge application. The bridge applies cleanly to the Paper 43 [47] hemispheric wedge at every finite cutoff. Seven newly accessible theorems follow (Theorems 6.2–6.8 + Corollary 6.9), headlined by **T6**: a **G2** *metric-level closure for the GeoVac wedge*, supplying a wedge-specific workaround for the published-open non-compact Latrémolière propinquity problem (Paper 45 [49] § 1.4 G2). The synthetic pLGH-convergence rate panel inherits the bit-exact numerical values $\{2.0746, 1.6101, 1.3223\}$ from Paper 45's Krein-side Λ^L. To the author's knowledge this is the first quantitative pLGH-convergence panel in the Mondino–Sämann literature derived from operator-algebraic input.

Honest scope. The bridge is K^+-weak-form only. The strong-form bridge extending $\mathbf{W}$ to the Paper 46 [50] Appendix B enlarged substrate with chirality-flipping generators (Q1') is deferred as a multi-month NCG-mathematics follow-on. The general G2 metric-level closure on non-wedge carriers remains the multi-month frontier; we close only the wedge-specific case via the bridge image on the synthetic side. The construction preserves $\mathbf{W}$'s

*Independent researcher. `jloutey@gmail.com`. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude. Implementation, internal memos, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility.

functorial (not isometric) character throughout: the Krein metametric $Ð^K$ and the synthetic time separation ℓ^L measure different physical quantities (thermal vs. geometric time) of the same wedge KMS state.

Place in the math.OA standalone series. Tenth in the GeoVac math.OA arc (siblings: Papers 29 [41], 32 [42], 38 [43], 39 [44], 40 [45], 42 [46], 43 [47], 44 [48], 45 [49], 46 [50], 47 [51]). The result composes Paper 45/46 verbatim on the Krein-algebraic side with the Mondino–Sämann synthetic-Lorentzian Gromov–Hausdorff framework on the synthetic side, mediated by the Connes–Rovelli thermal-time identification.

Contents

1 Introduction

1.1 Motivation: the F2 forward-vs-reverse triangle mismatch

Two recent advances in the convergence theory of generalized Lorentzian geometric structures, both published in December 2025, provide complementary but *a priori* incompatible frameworks:

- **Latrémolière's pinned proper quantum metric space framework** [25] (arXiv:2512.03573, December 3, 2025) extends the Latrémolière propinquity [22, 24] from quantum *compact* metric spaces to a quantum Gromov–Hausdorff *hypertopology* on the class of pointed proper quantum metric spaces. The resulting metametric $Ð^K$ satisfies a 4-point *relaxed forward* triangle inequality with codomain $[0, \infty]$. The framework is operator-algebraic: objects are C*-algebras with Lipschitz seminorms, topographies (Abelian C*-subalgebras), and pin states; morphisms are topographic quantum M-isometries.

- **Mondino–Sämann's covered Lorentzian pre-length space framework** [30] (arXiv:2504.10380 v4, December 9, 2025) extends the synthetic Lorentzian geometry of Kunzinger–Sämann pre-length spaces [19] to a pointed Lorentzian Gromov–Hausdorff (pLGH) convergence theory on covered LPLS. The synthetic time separation ℓ satisfies the 3-point *reverse* triangle inequality (causal-curve super-additivity of proper time) with codomain $\{-\infty\} \cup [0, \infty]$. The framework is synthetic-geometric: objects are sets with timelike-causal pre-orders, cover slabs, and basepoint events.

The two frameworks share many structural elements: pinning, exhaustive covers, slab bounds, ε-nets, pre-compactness, Gromov–Hausdorff convergence. They are aimed at different aspects of the same physics: generalized-geometric approximation of Lorentzian objects. However, the *algebraic shapes* of their central inequalities are incompatible:

	Latrémolière 2512.03573	Mondino–Sämann 2504.10380
codomain	$[0, \infty]$	$\{-\infty\} \cup [0, \infty]$
triangle direction	forward	reverse
triangle structure	4-point relaxed	3-point
sign convention	state-distance (≥ 0)	proper time (≥ 0 when causal)

We call this the **F2 mismatch**. Naive attempts at a bridge fail: negating the time variable (R1) puts the codomain in the wrong half-line and the 4-point structure does not collapse to 3-point under sign reversal. The substantive question is whether a bridge can be *categorical*: a functor sending Krein-algebraic objects (where relevant) to synthetic-Lorentzian objects (where relevant), with the F2 mismatch interpreted as a structural signature of two distinct projection mechanisms applied to the same physical object, rather than as a defect to be reconciled.

1.2 Main result: the Wick-rotation functor and the Krein–Mondino–Sämann Bridge

We construct (Definition 5.3, Theorem 5.4) the Wick-rotation functor

$$\mathbf{W} : \mathbf{KreinMetaMet}_{\mathrm{pp}} \longrightarrow \mathbf{LorPLG}_{\mathrm{cov}},$$

contravariant by Gelfand duality, sending a Krein-pointed proper QMS $(\mathcal{A}^K, L^K, \mathcal{M}^L, \omega^L_W)$ to the covered LPLS

$$\mathbf{W}(\mathcal{A}^K, L^K, \mathcal{M}^L, \omega^L_W) \;=\; (\hat{\mathcal{M}}^L,\; \ell^L,\; \hat{\omega}^L_W,\; \hat{\mathcal{U}}^L),$$

where $\hat{\mathcal{M}}^L$ is the Gelfand spectrum of the M-diagonal Abelian topography, $\ell^L = \kappa_g \cdot \tau^{\omega^L_W}_{\mathrm{mod}}$ is κ_g times the modular-flow time on the wedge boost orbits ($-\infty$ off-orbit), $\hat{\omega}^L_W$ is the Bisognano–Wichmann vacuum as character of the topography, and $\hat{\mathcal{U}}^L$ is the truncation cover.

The Krein–Mondino–Sämann Bridge Theorem (Theorem 5.4) states that all four bridge properties hold at theorem-grade rigor:

(B1) *Structural correspondence* — 15 rows of structural alignment (set, topology, time separation, basepoint, cover, scales, slabs, ε-nets, convergence, pre-compactness, etc.).

(B2) *Reverse triangle inequality* on the bridge image — on-orbit via additivity of Tomita–Takesaki modular flow at integer K^W_α spectrum [46]; off-orbit *structurally vacuous* at the K^+-weak-form bridge because the M-diagonal topography forces orbit-pair contradiction (Decomposition O Case (iii) is empty).

(B3) *Pre-compactness inheritance* — Mondino–Sämann Thm 6.2 applies on the bridge image via the Paper 44 propagation-number = 2 cardinality bound.

(B4) *Convergence transport* — Krein-side propinquity convergence under the Latrémolière hypertopology induces Mondino–Sämann pLGH convergence, with Plancherel-nested Berezin maps matching the MS Def 3.6 axioms one-to-one.

The F2 mismatch is resolved via the Connes–Rovelli thermal-time hypothesis [9]: the Krein metametric $Ð^K$ measures *thermal time* of the wedge KMS state, the synthetic time separation ℓ^L measures *geometric proper time*, and the Wick rotation between them is the analytic continuation mapping additive thermal-time structure to super-additive geometric-time structure. Concretely, on the Bisognano–Wichmann wedge, $t_{\mathrm{geometric}} = \kappa_g \cdot t_{\mathrm{thermal}}$ by the four-witness Wick-rotation theorem of Paper 42 [46].

1.3 GeoVac wedge application: seven newly accessible theorems

The bridge is illustrated on the canonical Paper 43 [47] hemispheric wedge $W_L = P_W^{\mathrm{spat}} \otimes P_{t\geq 0}$ on the compact-temporal Lorentzian Krein space $\mathcal{K}_{n_{\max},N_t,T}$, with Bisognano–Wichmann vacuum $\omega_W^L = e^{-K_\alpha^W}/Z$ and modular Hamiltonian $K_\alpha^W = J_{\mathrm{polar}}^W$ of integer spectrum two_m$_j$. All five Krein PPQMS axioms verify by direct construction. Seven newly accessible theorems follow (Section 6):

- **T1** (wedge as MS-covered LPLS) — every finite cutoff produces a well-defined Mondino–Sämann covered Lorentzian pre-length space with uniform timelike diameter 2π across all cover levels;
- **T2** (synthetic compactness of wedge family) — pLGH pre-compactness for sequences of GeoVac wedge cutoffs;
- **T3** (synthetic bit-exact panel) — the synthetic pLGH-rate panel at $(n_{\max}, N_t) \in \{(2,3),(3,5),(4,7)\}$ inherits bit-identically the Paper 45 Krein-side Λ^L values $\{2.0746, 1.6101, 1.3223\}$, the first quantitative pLGH-convergence panel in the MS literature derived from operator-algebraic input;
- **T4** (synthetic four-witness readings) — the Paper 42 [46] four-witness Wick-rotation theorem's six physical witnesses reduce to two synthetic-side equivalence classes distinguished by the rapidity-rate normalization κ_g;
- **T5** (synthetic Pythagorean orthogonality with M1 signature) — the operator-algebraic Pythagorean orthogonality $\langle H_{\mathrm{local}}, D_W^L\rangle_{HS} = 0$ of Paper 43 [47] § 10.2 transports to a synthetic-side foliation orthogonality, with the closed-form $1/\pi^2$ residual prefactor identified as the master Mellin engine M1 Hopf-base-measure signature (Paper 32 [42] § VIII);
- **T6** (**G2 metric-equivalent closure for the GeoVac wedge**) — the substantive deepest content: the de-compactification limit $T \to \infty$ closes at the pLGH-equivalent (metric-level-equivalent) level on the synthetic side for the GeoVac wedge specifically. A workaround for the published-open non-compact propinquity problem (Paper 45 [49] § 1.4 G2) at the GeoVac-wedge-specific level;
- **T6a** (synthetic three-carrier identification) — Paper 47 [51] § 6 three-carrier identification (periodic / Dirichlet / non-compact) extends from operator-algebraic to synthetic-Lorentzian level.

1.4 Honest scope

The bridge **W** is constructed at the **K$^+$-weak-form** level only (the substrate of Paper 45 [49]). The strong-form extension of **W** to the Paper 46 [50] Appendix B enlarged substrate (chirality-flipping generators M^{flip} with $\{J, M^{\mathrm{flip}}\} = 0$) is **not** constructed here; it is the named open question **Q1'** (Section 8.1). On the enlarged substrate the off-orbit super-additivity ceases to be vacuous and becomes a load-bearing operator-algebraic statement, turning Q1' into a substantive multi-month NCG-mathematics target.

The functor **W** is **functorial, not isometric**: $Ð^K$ on the Krein-algebraic side and ℓ^L on the synthetic side measure different physical quantities of the same wedge KMS state. The forward-vs-reverse triangle direction is not a "mismatch to be reconciled" but a structural signature of two distinct projection mechanisms applied to the same physical object. This is the F2 resolution.

For the GeoVac wedge, the substantive **T6** closure of G2 is wedge-specific: it closes the de-compactification limit at the propinquity-equivalent (pLGH) level on the synthetic side for the wedge specifically. The general non-compact Latrémolière propinquity problem (Paper 45 [49] § 1.4 G2 on arbitrary non-wedge carriers) remains the multi-month NCG-mathematics frontier and is **not** closed by this paper. The Paper 47 [51] norm-resolvent closure at the spectral level remains the operator-algebraic gold standard for general carriers; T6 specializes the metric-level question to the wedge via the bridge.

1.5 Roadmap of the paper

Section 2 recalls the necessary background: Latrémolière's pinned proper QMS framework (§ 2.1), the Krein operator-system substrate of Papers 44/45/46 (§ 2.2), the Mondino–Sämann pre-length space framework (§ 2.3), the Bisognano–Wichmann vacuum and the four-witness Wick-rotation theorem of Paper 42 (§ 2.4), and the Connes–Rovelli thermal-time hypothesis (§ 2.5).

Section 3 establishes the Krein-pointed proper quantum metric space: a 4-tuple $\mathbb{X}^K = (\mathcal{A}^K, L^K, \mathcal{M}^L, \omega_W^L)$ in which all nine Latrémolière axioms transport at theorem-grade rigor.

Section 4 constructs the Krein hypertopology on the class of Krein-pointed proper QMS, with explicit M-tunnel extent, M-local metametric, 4-point relaxed triangle inequality, coincidence theorem, and inframetric structure. The compact-case agreement at $N_t = 1$ reduces bit-exactly to the Paper 38 [43] SU(2) Riemannian propinquity hypertopology.

Section 5 constructs the Wick-rotation functor and states the Bridge Theorem. The F2 mismatch is resolved via R3 (Connes–Rovelli thermal time) as the unifying framework. The candidate resolutions R1 (negate time) and R2 (sub-functors) are tested and found insufficient.

Section 6 applies the bridge to the GeoVac hemispheric wedge, with the seven newly accessible theorems.

Section 7 states the synthetic-Lorentzian compactness theorems, with T6 (G2 metric-level closure for the GeoVac wedge) as the headline.

Section 8 catalogues the open questions: Q1' strong-form bridge with enlarged substrate, G2 on the general non-wedge carrier, G3 cross-manifold, G4 inner-factor calibration.

1.6 Related work

The Latrémolière lineage [11, 12, 21–25] provides the operator-algebraic side of the bridge. The pre-2025 papers handle the compact case; the 2512.03573 hypertopology paper [25] is the first to extend to proper (locally compact, non-unital) quantum metric spaces. Our Krein lift inherits and extends this framework.

The Mondino–Sämann lineage [19, 29, 30] provides the synthetic-Lorentzian side. The 2504.10380 v4 paper [30] introduces the pointed extension with covered LPLS and pLGH convergence; earlier papers by Kunzinger–Sämann established the underlying Lorentzian pre-length-space framework. The Mondino–Ryborz–Sämann stability paper [29] (May 2026) is the most recent flagship in this lineage.

Adjacent synthetic-Lorentzian convergence frameworks include: Sormani–Vega null distance [36], Sakovich–Sormani timed Gromov–Hausdorff [34], Minguzzi–Suhr bounded Lorentzian metric spaces [28], Müller's Cauchy slabs [31], Allen–Burtscher metric GH [1], Che–Perales–Sormani timed Hausdorff [7]. These frameworks operate in distinct categories from the present bridge (Mondino–Sämann's ε-net-of-causal-diamonds approach versus null-distance metric-GH or timed

Hausdorff variants); they are cited for completeness but are not in direct competition with the bridge construction.

The Connes–Rovelli thermal-time hypothesis [9] supplies the unifying interpretation. Active research on Connes-spectral-triples plus Tomita–Takesaki modular structure as a NCG framework for general relativity (e.g., Bertozzini–Conti–Lewkeeratiyutkul [2]) is adjacent to R3 but does not target the specific Krein-pointed PPQMS $\to$ Mondino–Sämann pLGH bridge constructed here.

Recent NCG-side work on twisted spectral triples and pseudo-Riemannian spectral geometry includes Bizi–Brouder–Besnard [4], van den Dungen [39], Franco–Eckstein [13], Strohmaier [37], and the recent Nieuviarts twisted-spectral-triple paper [32] (December 2025 v2). The Nieuviarts framework is restricted to even KO-dimension and does not apply directly to our KO-dim 3 substrate; we cite it as an alternative-to-Wick-rotation route in the Lorentzian-extension program. Ketterer [17] (May 2026) and Kubota [18] (May 2026) are recent synthetic-side internal refinements; the former extends ℓ-convergence on metric measure spaces, the latter establishes a Lorentzian coarea inequality. We cite both as adjacent works in the synthetic-Lorentzian convergence literature.

The Hekkelman–McDonald papers [15, 16] on Connes spectral triples and noncommutative integration provide non-compact propinquity machinery that does not extend to a metric-level closure on the $S^3 \times \mathbb{R}_t$ carrier; our T6 closure for the wedge sits on the synthetic side via the bridge rather than via direct non-compact propinquity.

To the author's knowledge, no published paper constructs a categorical bridge between operator-algebraic Lorentzian propinquity and synthetic Lorentzian Gromov–Hausdorff frameworks. The bridge constructed here is novel.

2 Setup

2.1 Latrémolière's pinned proper QMS framework

We recall the central definitions from Latrémolière [25].

Definition 2.1 (Latrémolière 2512.03573 Def 1.18: Leibniz seminorm)**.** Let $\mathcal{A}$ be a C*-algebra. A *Leibniz Lipschitz seminorm* L on $\mathcal{A}$ is a (typically unbounded) seminorm on a dense *-subalgebra $\mathrm{dom}(L) \subseteq \mathcal{A}$ satisfying $L(a^*) = L(a)$ and the Leibniz inequality $L(ab) \leq \|a\|L(b) + L(a)\|b\|$.

Definition 2.2 (Latrémolière 2512.03573 Def 1.22: separable QLCMS)**.** A *separable quantum locally compact metric space* (QLCMS) is a pair $(\mathcal{A}, L)$ of a separable C*-algebra $\mathcal{A}$ and a Leibniz seminorm L satisfying:

(FM) Fortet–Mourier distance metrizes the weak-$*$ topology on $\mathcal{S}(\mathcal{A})$;

(CB) the unit ball $\{a \in \mathrm{dom}(L) : L(a) \leq 1\}$ is closed;

(B) there exists a strictly positive $h \in \mathcal{A}_{\mathrm{sa}}$ and a state $\mu \in \mathcal{S}(\mathcal{A})$ with $\mu(h) = 1$ such that

$$\{hah : a = b + t\mathbf{1},\ L(b) \leq 1,\ \mu(b) = -t\}$$

is bounded.

Definition 2.3 (Latrémolière 2512.03573 Def 1.26: pinned QLCMS)**.** A *pinned QLCMS* is a triple $(\mathcal{A}, L, \mu)$ extending Definition 2.2 with $\mu \in \mathcal{S}(\mathcal{A})$ a pin state, such that $\{\phi \in \mathcal{S}(\mathcal{A}) : \mathrm{mk}_L(\phi, \mu) < \infty\}$ is weak-$*$ dense in $\mathcal{S}(\mathcal{A})$, where mk_L is the Monge–Kantorovich distance.

Lemma 2.4 (Latrémolière 2512.03573 Lemma 1.25)**.** *If $(\mathcal{A}, L)$ is a separable QLCMS with the (B)-axiom witnessed by (h, μ), then $\{\phi : \mathrm{mk}_L(\phi, \mu) < \infty\}$ is weak-$*$ dense in $\mathcal{S}(\mathcal{A})$. In particular, the (B)-axiom of Definition 2.2 implies the density condition of Definition 2.3.*

Definition 2.5 (Latrémolière 2512.03573 Def 1.37 + 1.42: pointed proper QMS)**.** A *pinned proper QMS* is a pinned QLCMS $(\mathcal{A}, L, \mu)$ containing an L-Lipschitz μ-pinned exhaustive sequence $(h_n) \subseteq \mathrm{dom}(L)$ with $L(h_n) \to 0$, $\mu(h_n) \to 1$, $\|h_n\| \to 1$. A *topography* $\mathfrak{M} \subseteq \mathcal{A}$ is an Abelian C*-subalgebra containing such an exhaustive sequence. A *pointed proper QMS* is a 4-tuple $(\mathcal{A}, L, \mathfrak{M}, \mu)$ with $\mathfrak{M}$ a topography and μ restricting to a character of $\mathfrak{M}$.

The associated structures are M-isometries, M-tunnels, M-local metametric $\eth_M$, parameter-family metametric $Đ^K$, and a 4-point relaxed forward triangle:

$$Đ^K(\mathbb{A}, \mathbb{B}) \leq (1 + Đ^K(\mathbb{A}, \mathbb{D}))^2 Đ^K(\mathbb{D}, \mathbb{B}) + (1 + Đ^K(\mathbb{D}, \mathbb{B}))^2 Đ^K(\mathbb{A}, \mathbb{D}).$$

The Latrémolière coincidence theorem (Thm 3.6) characterizes $Đ^K = 0$ via the existence of a full topographic M-isometry.

2.2 Krein operator-system substrate from Papers 44/45/46

The Krein-side substrate is the compact-temporal Lorentzian Krein space

$$\mathcal{K}_{n_{\max}, N_t, T} = \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$$

of Paper 44 [48], with fundamental symmetry $J_L = J_{\mathrm{spat}} \otimes I_{N_t}$ where $J_{\mathrm{spat}}^2 = +I$ is the spatial chirality swap; the K$^+$-restricted Hilbert space is $\mathcal{K}^+ = \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$. The Lorentzian Dirac is (van den Dungen 2016 Prop. 4.1 [39])

$$D_L = i\big(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t}\big),$$

Krein-self-adjoint: $D_L^\times = D_L$ where $\dagger$ is the Krein adjoint $\gamma^0 \cdot * \cdot \gamma^0$. The natural Lorentzian operator system is

$$\mathcal{O}^L = \mathrm{span}_{\mathbb{C}}\{M^{\mathrm{spat}}_{N,L,M} \otimes M^{\mathrm{temp}}_p : \ N \leq n_{\max},\ L < N,\ |M| \leq L,\ 0 \leq p \leq N_t - 1\},$$

of finite dimension at finite cutoff, with propagation number $\mathrm{prop}(\mathcal{O}^L) = 2$ under the Weyl-doubled achievable envelope (Paper 44 [48] Prop. prop=2).

Paper 45 [49] establishes the K$^+$-restricted weak-form Lorentzian propinquity convergence:

$$\Lambda^L(\mathcal{T}^L_{n_{\max}, N_t, T}, \mathcal{T}^L_{S^3 \times S^1_T}) \ \leq\ C_3^{\mathrm{joint}} \cdot \gamma^{\mathrm{joint}}_{n_{\max}, N_t, T} \ \to\ 0$$

along admissible scaling $(n_{\max}, N_t) \to (\infty, \infty)$ at fixed T, with bit-exact panel values $\Lambda^L(2,3) = 2.0746$, $\Lambda^L(3,5) = 1.6101$, $\Lambda^L(4,7) = 1.3223$. Paper 46 [50] extends to a strong-form (no K$^+$ restriction) on the natural chirality-doubled scalar-multiplier substrate, with bit-identical $\Lambda^{\mathrm{strong}} = \Lambda^L$ ("free upgrade"), and to a further enlarged substrate (Appendix B) with chirality-flipping generators M^{flip} satisfying $\{J_L, M^{\mathrm{flip}}\} = 0$.

2.3 Mondino–Sämann pre-length space framework

We recall the central definitions from Mondino–Sämann [30].

Definition 2.6 (Mondino–Sämann Def 2.3: Lorentzian pre-length space)**.** A *Lorentzian pre-length space* (LPLS) is a pair (X, ℓ) with X a set carrying a topology finer than the chronological topology, and $\ell : X \times X \to \{-\infty\} \cup [0, \infty]$ satisfying $\ell(x, x) \geq 0$ and the *reverse triangle inequality*

$$\ell(x, y) + \ell(y, z) \leq \ell(x, z) \quad \forall x, y, z \in X \tag{1}$$

(with convention $\pm\infty + \mp\infty = 0$ on the LHS yielding $-\infty$ when undefined). Timelike, causal, and causal-diamond relations are derived in the standard way: $x \ll y \iff \ell(x, y) > 0$, $x \leq y \iff \ell(x, y) \geq 0$, $J(x, y) = \{z : x \leq z \leq y\}$, and the time separation $\tau := \max(0, \ell)$.

Definition 2.7 (Mondino–Sämann Def 3.2: ε-net)**.** An *ε-net* for $A \subseteq X$ is a collection $\mathcal{S} = (J_i)_{i\in\Omega}$ of causal diamonds with $\tau(J_i) \le \varepsilon$ and $A \subseteq \bigcup_i J_i$.

Definition 2.8 (Mondino–Sämann Def 3.6: LGH convergence of subsets)**.** A sequence $A_n \to A$ converges in the Lorentzian Gromov–Hausdorff (LGH) sense if there exist ε-nets S_n for A_n and S for A with: (i) matching cardinality; (ii) correspondences R_n between vertices with $\mathrm{dis}(R_n) \to 0$; (iii) extension property of correspondences; (iv) forward density.

Definition 2.9 (Mondino–Sämann Def 3.8: covered LPLS)**.** A *covered LPLS* is a 4-tuple $(X, \ell, o, \mathcal{U})$ where (X, ℓ) is an LPLS, $o \in X$ is a basepoint event, and $\mathcal{U} = (U_k)_{k\in\mathbb{N}}$ is a countable cover satisfying: (i) $\bigcup_k U_k = X$; (ii) $U_k \subseteq U_{k+1}$ (nested); (iii) $o \in U_k$ for all k; (iv) $\sup_{x,y\in U_k} \tau(x, y) < \infty$ (slab bounds).

Definition 2.10 (Mondino–Sämann Def 3.12: pLGH convergence)**.** $(X_n, \ell_n, o_n, \mathcal{U}_n) \to (X, \ell, o, \mathcal{U})$ in the *pointed LGH* sense iff for each k, $U_{k,n} \to U_{k,\infty}$ in LGH sense.

The Mondino–Sämann pre-compactness theorem (Thm 6.2) gives that any sequence of covered LPLS with (i) uniform slab bounds, (ii) uniform ε-net cardinality bounds, (iii) cumulative ε-net nesting has a pLGH-convergent subsequence.

2.4 Bisognano–Wichmann vacuum and the four-witness Wick-rotation theorem

The Paper 43 [47] hemispheric wedge is

$$W_L := P_W^{\mathrm{spat}} \otimes P_{t\ge 0}, \quad P_W^{\mathrm{spat}} = \tfrac{1}{2}(I + R_{\mathrm{polar}}),$$

where R_{polar} is the spatial m_j-reflection involution (Paper 42 [46] § 4) and $P_{t\ge0}$ is the diagonal projection on $\mathbb{C}^{N_t}$ to positive temporal indices. The Bisognano–Wichmann vacuum state under the BW-α geometric Hamiltonian (with $\beta = 2\pi/\kappa_g$) is

$$\rho_W^L := e^{-K_\alpha^W}/Z, \quad K_\alpha^W = J_{\mathrm{polar}}^W \text{ on the wedge},$$

with integer spectrum $\mathrm{two\_m}_j \in \mathbb{Z}_{>0}$. The Tomita–Takesaki modular Hamiltonian $K_{\mathrm{TT}} = -\log\Delta$ is constructed from the wedge KMS state via polar decomposition (Paper 42 [46] § 6).

The Paper 42 [46] four-witness Wick-rotation theorem states that six physical witnesses (Bisognano–Wichmann canonical with $\kappa_g = 1$, Hartle–Hawking at $M = 1, 2$, Sewell at $M = 1$, Unruh at $a = 1, 2$) all yield bit-identical modular operators Δ and Hamiltonians K_{TT} under the BW unit normalization $H_{\mathrm{local}} := K_\alpha^W/\beta$. The Riemannian-side bit-exact identity $\sigma_{2\pi}^\alpha(O) = O$ holds at every finite cutoff $n_{\max} \in \{1, 2, 3, 4, 5\}$, and the Tomita BW-$\gamma$ flow σ_t^{TT} satisfies the conjugacy $\sigma_t^{TT}(a) = \sigma_{-t}^\alpha(a)$ bit-exact. Paper 43 [47] extends to Lorentzian signature via BBB Krein construction [4] and verifies the same six-witness collapse at every $(n_{\max}, N_t)$ panel cell.

2.5 Connes–Rovelli thermal-time hypothesis

Connes and Rovelli [9] (1994) proposed:

Thermal-time hypothesis. *In quantum statistical mechanics on a generally covariant system, the natural physical time evolution at thermal equilibrium with state ω is given by the modular automorphism group σ_t^ω of ω (a state on the observable algebra $\mathcal{A}$).*

Applied to a KMS state at inverse temperature β on a Type III von Neumann algebra arising as the local algebra of a region of spacetime (Bisognano–Wichmann setting), the thermal time of ω relates to the geometric proper time of the underlying spacetime by

$$t_{\mathrm{thermal}} = \frac{\beta}{2\pi} \cdot t_{\mathrm{geometric}} = \frac{1}{\kappa_g} \cdot t_{\mathrm{geometric}}, \tag{2}$$

where κ_g is the surface gravity (equivalently the BW Hawking-temperature parameter). This is the four-witness Wick-rotation theorem of Paper 42 [46] at the kinematic level: modular flow IS Lorentz-boost orbit on the BW wedge, with proportionality factor $1/\kappa_g = \beta/(2\pi)$.

2.6 Notation and conventions

Throughout the paper: $\mathcal{K} = \mathcal{K}_{n_{\max},N_t,T} = \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ denotes the compact-temporal Krein space; $J_L = J_{\mathrm{spat}} \otimes I_{N_t}$ the fundamental symmetry; $\mathcal{K}^+, \mathcal{K}^-$ the K$^\pm$-eigenspaces of J_L; $\mathcal{O}^L$ the natural Lorentzian operator system; D_L the Krein-self-adjoint Lorentzian Dirac; $\omega_W^L = e^{-K_\alpha^W}/Z$ the BW vacuum; $K_\alpha^W = J_{\mathrm{polar}}^W$ the BW geometric Hamiltonian (with integer spectrum two_m$_j$); $\sigma_t^{\omega_W^L}$ the Tomita–Takesaki modular flow of ω_W^L; $\tau_{\mathrm{mod}}^{\omega_W^L}(\hat{\omega}, \hat{\omega}') := \inf\{t \geq 0 : \hat{\omega}' = \hat{\omega} \circ \sigma_t^{\omega_W^L}\}$ the modular-flow time.

We write $\mathrm{Spec}(\mathfrak{B})$ for the Gelfand spectrum of an Abelian C*-algebra $\mathfrak{B}$ and $\hat{\omega}$ for the character induced by a state ω restricting to $\mathfrak{B}$.

3 Krein-pointed proper quantum metric spaces

We lift Latrémolière's pinned proper QMS framework to the Krein-algebraic setting. Throughout this section, the substrate is the K$^+$-restricted Hilbert space; the natural-substrate generators all commute with J_L, so the Krein operator norm coincides with the Hilbert operator norm on the K$^+$ restriction.

3.1 The Krein-Leibniz Lipschitz seminorm

Definition 3.1 (Krein-Leibniz Lipschitz seminorm)**.** On the natural Lorentzian operator system $\mathcal{O}^L$ (or its continuum limit on $\mathcal{T}^L_{S^3 \times S^1_T}$), define

$$L^K(a) := \|[D_L, a]\|_{\mathrm{op}}, \qquad a \in \mathrm{dom}(L^K) \subseteq \mathcal{O}^L,$$

where $\|\cdot\|_{\mathrm{op}}$ is the operator norm on $\mathcal{B}(\mathcal{K})$ restricted to $\mathcal{K}^+$. We call L^K the *Krein-Leibniz Lipschitz seminorm.*

Lemma 3.2 (Krein-Leibniz inequality)**.** *For every a, b in the natural chirality-doubled scalar-multiplier substrate,*

$$L^K(ab) \leq \|a\|_{\mathcal{O}^L} L^K(b) + L^K(a)\|b\|_{\mathcal{O}^L}.$$

Proof. On the natural substrate, all elements of $\mathcal{O}^L$ commute with J_L [48], so both a, b restrict cleanly to $\mathcal{K}^+$. On $\mathcal{K}^+$, the standard Hilbert operator norm $\|\cdot\|_{\mathrm{op}}|_{\mathcal{K}^+}$ agrees with the algebra norm $\|\cdot\|_{\mathcal{O}^L}$. The standard operator-algebra Leibniz identity $[D_L, ab] = [D_L, a]b + a[D_L, b]$ holds in $\mathcal{B}(\mathcal{K})$; restricting to $\mathcal{K}^+$ and applying submultiplicativity yields the Krein-Leibniz inequality. □

Remark 3.3 (Krein vs Hilbert operator norm on the enlarged substrate)**.** The natural-substrate version of Lemma 3.2 uses the Hilbert operator norm on $\mathcal{K}^+$. On the Paper 46 [50] Appendix B *enlarged* substrate with chirality-flipping generators M^{flip} satisfying $\{J_L, M^{\mathrm{flip}}\} = 0$, the K$^+$ restriction is no longer well-defined as a multiplier (such generators map $\mathcal{K}^+$ to $\mathcal{K}^-$), and the two natural operator norms (Krein operator norm with respect to the indefinite Krein product vs. Hilbert operator norm on the full Krein space) differ. The K$^+$-weak-form route of this paper makes the distinction invisible; the strong-form route makes it visible at the level of the gradient-norm absorption mechanism of Paper 46 [50] Appendix B. For Q1' (Section 8.1) this distinction becomes load-bearing.

3.2 The Krein-separable QLCMS and Krein-pinned QLCMS

Definition 3.4 (Krein-separable QLCMS)**.** A *Krein-separable QLCMS* is a pair $(\mathcal{A}^K, L^K)$ where $\mathcal{A}^K$ is the natural Lorentzian operator system on $\mathcal{K}$ (at finite cutoff: the C*-algebra generated by $\mathcal{O}^L$ in $\mathcal{B}(\mathcal{K}^+)$, a finite-dimensional matrix algebra; in the continuum: $\mathcal{A}^K = C(\mathcal{T}^L_{S^3\times S^1_T})$ acting on $\mathcal{K}^+_{\text{continuum}}$), and L^K is the Krein-Leibniz Lipschitz seminorm on a dense *-subalgebra $\text{dom}(L^K) \subseteq \mathcal{A}^K$, satisfying:

(FM-K) The Krein-positive Fortet–Mourier distance metrizes the weak-$*$ topology on $\mathcal{S}(\mathcal{A}^K)$;

(CB-K) The unit ball $\{a \in \text{dom}(L^K) : L^K(a) \leq 1\}$ is closed in the operator-norm topology;

(B-K) There exists strictly positive $h^L \in \mathcal{A}^K_{\text{sa}}$ and a state $\mu^L \in \mathcal{S}(\mathcal{A}^K)$ with $\mu^L(h^L) = 1$ such that

$$\{h^L a h^L : a = b + t\mathbf{1},\ L^K(b) \leq 1,\ \mu^L(b) = -t\}$$

is bounded.

Lemma 3.5. *The pair $(\mathcal{A}^K, L^K)$ with the natural Lorentzian operator system substrate is a Krein-separable QLCMS, with witnessing $(h^L, \mu^L) = (K^W_\alpha/Z, \omega^L_W)$.*

Proof sketch. The (FM-K), (CB-K), (B-K) axioms transport from Papers 42–45 substrate: (FM-K) via Paper 44 [48] § 6 Krein-positive Wasserstein–Kantorovich SDP framework, with weak-$*$ metrization following Latrémolière [25] Thm 1.9; (CB-K) via Banach–Alaoglu plus lower-semicontinuity of L^K; (B-K) via Paper 42 [46] § 5 integer spectrum of K^W_α (h^L positive definite, $\|h^L\| = 1$, $\omega^L_W(h^L) = 1$ by BW pinning) combined with Paper 44 [48] propagation number $= 2$ (bounded sets in finite-dimensional subspaces are bounded). See A.2' § 2.3 of the synthesis memo for full detail. □

Definition 3.6 (Krein-pinned QLCMS)**.** A *Krein-pinned QLCMS* is a triple $(\mathcal{A}^K, L^K, \omega^L_W)$ with $\omega^L_W \in \mathcal{S}(\mathcal{A}^K)$ the BW vacuum, such that $\{\phi \in \mathcal{S}(\mathcal{A}^K) : \text{mk}_{L^K}(\phi, \omega^L_W) < \infty\}$ is weak-$*$ dense in $\mathcal{S}(\mathcal{A}^K)$.

Lemma 3.7 (Continuum tightness via Lemma 2.4)**.** *$(\mathcal{A}^K, L^K, \omega^L_W)$ is a Krein-pinned QLCMS. Equivalently: the density condition of Definition 3.6 holds.*

Proof. By Lemma 3.5, the (B-K) axiom holds with $(h^L, \mu^L) = (K^W_\alpha/Z, \omega^L_W)$. By Lemma 2.4 (Latrémolière 2512.03573 Lemma 1.25), the density of finite-Monge–Kantorovich states follows mechanically. □

Remark 3.8. Lemma 3.7 dissolves a caveat that the diagnostic phase of the program had flagged as a 2–3 week sub-sprint: the continuum tightness condition is a *corollary* of the (B-K) boundedness axiom via Latrémolière's own Lemma 1.25. The Krein lift requires no separate verification beyond the (B-K) ingredients.

3.3 The Krein L-Lipschitz ω^L_W-pinned exhaustive sequence and the approximate-unit theorem

Definition 3.9 (Krein L-Lipschitz ω^L_W-pinned exhaustive sequence)**.** A *Krein L-Lipschitz ω^L_W-pinned exhaustive sequence* is a sequence $(h_n)_{n\in\mathbb{N}} \subseteq \text{dom}_{\text{sa}}(L^K)$ with:

1. $\lim_{n\to\infty} L^K(h_n) = 0$,
2. $\lim_{n\to\infty} \omega^L_W(h_n) = 1$,
3. $\lim_{n\to\infty} \|h_n\|_{\mathcal{A}^K} = 1$.

Theorem 3.10 (Krein approximate-unit, transport of Latrémolière 2512.03573 Thm 1.30). *If $(\mathcal{A}^K, L^K, \omega_W^L)$ is a Krein-pinned QLCMS and (h_n) is a Krein L-Lipschitz ω_W^L-pinned exhaustive sequence, then (h_n) is an approximate unit for $\mathcal{A}^K$.*

Proof sketch. Direct transport of Latrémolière 2512.03573 Thm 1.30 via Claims 1.31–1.34: weak-$*$ convergence to 1 (using Lemma 3.7); h_n^2 pinning (via Lemma 3.2 Leibniz applied to $h_n \cdot h_n$); $(1 - h_n^2)$ vanishing on compact sets of states (via Arzelà–Ascoli on the Fortet–Mourier metric ball); convergence to 1 in strict topology (via Kadison functional calculus on the K^+-restricted commutative envelope of h_n). □

The explicit Krein-side exhaustive sequence. The natural Krein-side exhaustive sequence is the truncated Krein triple projector sequence:

$$h_n := P_{n_{\max}(n), N_t(n), T(n)},$$

the orthogonal projection onto the $\mathcal{H}_{\mathrm{GV}}^{n_{\max}(n)} \otimes \mathbb{C}^{N_t(n)}$ truncation, with $(n_{\max}(n), N_t(n), T(n)) \to (\infty, \infty, T_\infty)$ along admissible scaling per Paper 47 [51]. Verification: (i) $L^K(h_n) \le \gamma^{\mathrm{joint}}_{n_{\max}(n), N_t(n), T(n)} \cdot O(1) \to 0$ by Paper 45 [49] Sub-Sprint C approximate-identity property; (ii) $\omega_W^L(h_n) \to 1$ by support of ω_W^L on the wedge; (iii) $\|h_n\| = 1$ since h_n is a projector.

3.4 Krein topography and Krein-pointed proper QMS

Definition 3.11 (Krein topography). A *Krein topography* $\mathcal{M}^L$ of a Krein-separable QLCMS $(\mathcal{A}^K, L^K)$ is an Abelian C*-subalgebra of $\mathcal{A}^K$ containing a strictly positive $h^L \in \mathrm{dom}(L^K)$ and admitting a character μ^L of $\mathcal{M}^L$ witnessing the (B-K) boundedness axiom restricted to $\mathcal{M}^L$.

Lemma 3.12 (The natural Krein topography). *The natural Krein topography on the GeoVac substrate is the sub-operator-system*

$$\mathcal{M}^L := \mathrm{span}_{\mathbb{C}}\big\{M^{\mathrm{spat}}_{N,L,0} \otimes I_{N_t} :\ N \le n_{\max},\ L < N\big\} \subseteq \mathcal{A}^K,$$

i.e. the chirality-doubled scalar multipliers with M-quantum-number $M = 0$ tensored with the temporal identity.

Proof sketch. Abelian: Each $M^{\mathrm{spat}}_{N,L,0}$ is a multiplication operator by a spatial 3-Y function $Y^{(3)}_{N,L,0}$, hence the algebra is Abelian; tensoring with I_{N_t} preserves the structure.

Strictly positive h^L: Take $h^L = K^W_\alpha / Z$; the BW geometric Hamiltonian is M-diagonal and positive on the wedge by Paper 42 [46] § 5.

(B-K) restricted to $\mathcal{M}^L$: Restrict Lemma 3.5 to $\mathcal{M}^L$; the boundedness condition is preserved.

Contains approximate unit: By Latrémolière Remark 1.41, this follows from the Leibniz property applied to the truncated projector sequence h_n, which is M-block-diagonal hence in $\mathcal{M}^L$. □

Definition 3.13 (Krein-pointed proper QMS). A *Krein-pointed proper quantum metric space* (Krein PPQMS) is a 4-tuple

$$\mathbb{X}^K = (\mathcal{A}^K, L^K, \mathcal{M}^L, \omega_W^L)$$

where $(\mathcal{A}^K, L^K, \omega_W^L)$ is a Krein-pinned QLCMS, $\mathcal{M}^L$ is a Krein topography of $(\mathcal{A}^K, L^K)$, ω_W^L restricts to a character of $\mathcal{M}^L$, and there exists a Krein L-Lipschitz ω_W^L-pinned exhaustive sequence contained in $\mathcal{M}^L$.

Lemma 3.14 (The merged substrate is a Krein PPQMS). *The 4-tuple $(\mathcal{A}^K, L^K, \mathcal{M}^L, \omega_W^L)$ with $\mathcal{A}^K$ the natural Lorentzian operator system, L^K the Krein-Leibniz Lipschitz seminorm, $\mathcal{M}^L$ the natural Krein topography of Lemma 3.12, and ω_W^L the BW vacuum, satisfies all axioms of Definition 3.13.*

Proof sketch. The (FM-K), (CB-K), (B-K) axioms hold by Lemma 3.5; the continuum tightness holds by Lemma 3.7; the topography axioms hold by Lemma 3.12; $\omega_W^L|_{\mathcal{M}^L}$ is a character because $\mathcal{M}^L$ is Abelian (states on Abelian C*-algebras are characters by Gelfand); the truncated projector sequence $h_n = P_{n_{\max}(n),N_t(n),T(n)}$ lies in $\mathcal{M}^L$ (M-block-diagonal) and is a Krein L-Lipschitz ω_W^L-pinned exhaustive sequence. □

3.5 Krein M-tunnels with extent element

Definition 3.15 (Krein M-tunnel)**.** Let $\mathbb{X}_i^K = (\mathcal{A}_i^K, L_i^K, \mathcal{M}_i^L, \mu_i)$ for $i = 1, 2$ be two Krein PPQMS and $M \geq 1$. A *Krein M-tunnel* from $\mathbb{X}_1^K$ to $\mathbb{X}_2^K$ is a 6-tuple

$$\tau^L = (\mathcal{D}^L, L_{\mathcal{D}^L}^K, \mathfrak{M}_{\mathcal{D}^L}^L, \pi_1^K, \pi_2^K, e^L)$$

where $(\mathcal{D}^L, L_{\mathcal{D}^L}^K, \mathfrak{M}_{\mathcal{D}^L}^L, \mu_1 \circ \pi_1^K)$ is a pointed topographic separable Krein-side QLCMS, $e^L \in \mathrm{dom}_{\mathrm{sa}}(L_{\mathcal{D}^L}^K) \cap \mathfrak{M}_{\mathcal{D}^L}^L$ is the *extent element*, and $\pi_i^K : \mathcal{D}^L \twoheadrightarrow \mathcal{A}_i^K$ are topographic Krein M-isometries.

The natural Krein M-tunnel. For comparison of the truncated Krein triple $\mathcal{T}_{n_{\max},N_t,T}^L$ with the continuum $\mathcal{T}_{S^3 \times S_T^1}^L$:

- $\mathcal{D}^L = \mathcal{O}_{n_{\max},N_t,T}^L \oplus C(\mathcal{T}_{S^3 \times S_T^1}^L)$, direct sum of truncated operator system and continuum function algebra;
- $L_{\mathcal{D}^L}^K((a, f)) = \max\{L_1^K(a), L_2^K(f), \|a - B^{\mathrm{joint}}(f)\|_{\mathrm{op}}/\varepsilon_0\}$, the tunnel Lipschitz combining intrinsic seminorms with Berezin discrepancy at scale $\varepsilon_0 = \gamma^{\mathrm{joint}}$;
- $\mathfrak{M}_{\mathcal{D}^L}^L = \mathcal{M}_1^L \oplus \mathcal{M}_2^L$;
- $\pi_1^K(a, f) = a$, $\pi_2^K(a, f) = f$;
- $e^L = (h_n, \mathbf{1})$, the natural extent element pairing the truncation projector with the continuum unit.

Lemma 3.16. *The natural Krein M-tunnel satisfies Definition 3.15.*

Proof sketch. The topography $\mathfrak{M}_{\mathcal{D}^L}^L$ is Abelian by direct sum of Abelian sub-algebras; e^L is in $\mathrm{dom}_{\mathrm{sa}}(L_{\mathcal{D}^L}^K)$ because both h_n and $\mathbf{1}$ are; $\pi_{1,2}^K$ are topographic M-isometries by Paper 45 [49] Sub-Sprint D Berezin/projection construction; e^L is K^+-positive because both factors commute with J_L trivially. □

4 Krein-pointed propinquity hypertopology

4.1 Extent of a Krein M-tunnel

Definition 4.1 (Krein extent of an M-tunnel)**.** The *Krein extent* of an M-tunnel $\tau^L = (\mathcal{D}^L, L_{\mathcal{D}^L}^K, \mathfrak{M}_{\mathcal{D}^L}^L, \pi_1^K, \pi_2^K, e$ is

$$\chi^K(\tau^L) := \max\{\chi_1^K, \chi_2^K, \chi_3^K\},$$

where χ_1^K is the reach (Hausdorff distance of pullback quasi-state spaces), χ_2^K is the height (pin-state alignment via Monge–Kantorovich on K^+ Wasserstein–Kantorovich), and χ_3^K is the extent-element Lipschitz-and-normalization:

$$\chi_3^K(\tau^L) = \max\{2ML^K(e^L),\ |1 - \mu_1 \circ \pi_1^K(e^L)|,\ |1 - \mu_2 \circ \pi_2^K(e^L)|\}.$$

Lemma 4.2. *For the natural Krein M-tunnel from $\mathcal{T}^L_{n_{\max},N_t,T}$ to $\mathcal{T}^L_{S^3\times S^1_T}$,*

$$\chi^K(\tau^L) \le \gamma^{\text{joint}}_{n_{\max},N_t,T} \to 0,$$

bit-identical to the Paper 45 [49] Sub-Sprint D propinquity bound.

Proof sketch. $\chi^K_1 = \text{reach}^K_B \le \gamma^{\text{joint}}$ by Paper 45 [49] Sub-Sprint D § 3.1; $\chi^K_2 = \text{K}^+$ pin-state distance, bounded by γ^{joint}; χ^K_3 has three components, each bounded by γ^{joint} times a constant via the approximate-identity property of h_n applied to the extent element. □

4.2 The Krein M-local quantum metametric

Definition 4.3 (Krein M-local quantum metametric)**.** For $M \ge 1$ and Krein PPQMS $\mathbb{X}^K, \mathbb{Y}^K$:

$$\eth^K_M(\mathbb{X}^K, \mathbb{Y}^K) := \inf\{\chi^K(\tau^L) : \ \tau^L \text{ is a Krein M-tunnel from } \mathbb{X}^K \text{ to } \mathbb{Y}^K\}.$$

Theorem 4.4 (4-point relaxed forward triangle inequality)**.** *For any $M \ge 1$ and Krein PPQMS $\mathbb{X}^K, \mathbb{Y}^K, \mathbb{Z}^K$:*

$$\eth^K_M(\mathbb{X}^K, \mathbb{Y}^K) \le (1 + \eth^K_M(\mathbb{X}^K, \mathbb{Z}^K))^2\, \eth^K_M(\mathbb{Z}^K, \mathbb{Y}^K) + (1 + \eth^K_M(\mathbb{Z}^K, \mathbb{Y}^K))^2\, \eth^K_M(\mathbb{X}^K, \mathbb{Z}^K).$$

Proof sketch. Direct transport of Latrémolière 2512.03573 Thm 2.24 via Krein-side tunnel composition. The natural-substrate Krein structure is closed under tunnel composition (direct sums of operator systems give operator systems, Berezin composition preserves K$^+$-positivity, the extent-element construction generalizes to triple composition). □

4.3 Coincidence

Theorem 4.5 (Coincidence)**.** *For Krein PPQMS $\mathbb{X}^K, \mathbb{Y}^K$, there exists a full topographic Krein M-isometry $\pi^K : \mathbb{X}^K \to \mathbb{Y}^K$ with $\mu_{\mathbb{Y}^K} \circ \pi^K = \mu_{\mathbb{X}^K}$ if and only if $\eth^K_M(\mathbb{X}^K, \mathbb{Y}^K) = 0$.*

Proof sketch. The "if" direction is contained in Theorem 4.4. The "only if" direction transports Latrémolière 2512.03573 Thm 3.6 via target-set machinery: given a sequence of tunnels τ^K_n with $\chi^K(\tau^K_n) \to 0$, the target sets $\mathfrak{t}^K_{\tau^K}(a|l) := \{\pi^K_2(d) : \pi^K_1(d) = a,\ \|d\|_{L_{\mathcal{D}^L},M} \le l\}$ are well-defined on the Krein side because of the K$^+$-positivity of natural-substrate generators, and the four almost-properties (Latrémolière Lemmas 3.2–3.5) transport mechanically via Krein-Leibniz and Krein-approximate-unit. The diagonal extraction with Banach–Alaoglu on the K$^+$-restricted state space produces a limit *-isomorphism preserving topography and BW pin state. □

4.4 The Krein quantum metametric and hypertopology

Definition 4.6 (Krein quantum metametric)**.** For Krein PPQMS $\mathbb{X}^K, \mathbb{Y}^K$:

$$Ð^K(\mathbb{X}^K, \mathbb{Y}^K) := \max\Big\{ \inf\{\varepsilon > 0 : \sup_{r\in[1,1/\varepsilon]} \eth^K_r(\mathbb{X}^K, \mathbb{Y}^K) < \varepsilon\},\ \left| e^{-\text{qdiam}(\mathbb{X}^K)} - e^{-\text{qdiam}(\mathbb{Y}^K)} \right| \Big\}.$$

Theorem 4.7 (Inframetric structure)**.** *$Ð^K$ is a K$^+$-positive inframetric: symmetric, satisfying the 4-point relaxed triangle, and $Ð^K(\mathbb{X}^K, \mathbb{Y}^K) = 0$ iff a full topographic Krein M-isometry exists between $\mathbb{X}^K$ and $\mathbb{Y}^K$.*

Proof sketch. Symmetry holds because Krein M-tunnels are symmetric in their roles; the 4-point relaxed triangle follows from Theorem 4.4; the coincidence statement is Theorem 4.5. □

Definition 4.8 (Krein hypertopology)**.** The *Krein hypertopology* on the class $p\mathcal{P}^K$ of Krein PPQMS is the topology with Kuratowski closure operator

$$\text{cl}^K(\mathcal{A}) := \{\mathbb{Y}^K \in p\mathcal{P}^K : Ð^K(\mathbb{Y}^K, \mathcal{A}) = 0\}$$

where $Ð^K(\mathbb{Y}^K, \mathcal{A}) := \inf\{Ð^K(\mathbb{Y}^K, \mathbb{X}^K) : \mathbb{X}^K \in \mathcal{A}\}$.

4.5 Compact-case agreement at $N_t = 1$

Theorem 4.9 (Compact-case agreement). *When restricted to $N_t = 1$, the Krein hypertopology on the class of Krein PPQMS reduces bit-exactly to the Paper 38 [43] SU(2) Riemannian propinquity hypertopology.*

Proof sketch. At $N_t = 1$: $\mathcal{K}_{n_{\max},1,T} = \mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ (spatial Krein space alone); $J_L = J_{\mathrm{spat}}$; $\mathcal{K}^+ = \mathcal{H}_{\mathrm{GV}}^{n_{\max},+}$; $D_L|_{N_t=1} = iD_{\mathrm{GV}}$ (temporal direction contributes nothing); $\mathcal{O}^L|_{N_t=1}$ is the chirality-doubled spinor lift of Paper 38 [43]'s scalar substrate. By Paper 45 [49] Sub-Sprint D § 5 Riemannian-limit recovery, $\Lambda^L_{n_{\max},1,T} = \Lambda^{\text{Paper 38}}_{n_{\max}}$ bit-exact. The pointed-propinquity reduction follows. □

Remark 4.10 (The compact-case agreement object). The compact-case object is the chirality-doubled spinor lift of Paper 38's Riemannian SU(2) Lipschitz spectral triple, pinned at the BW vacuum on the spatial wedge. This explicit identification ensures the Krein-pointed propinquity hypertopology is a strict extension of Paper 38, not a parallel construction.

4.6 Numerical panel

The Krein-side propinquity bound inherits the bit-exact numerical panel from Paper 45 [49]:

$(n_{\max}, N_t)$	$ð^K_M$ (qualitative)	Λ^L (Paper 45)
$(2,3)$	≤ 2.0746	$= 2.0746$
$(3,5)$	≤ 1.6101	$= 1.6101$
$(4,7)$	≤ 1.3223	$= 1.3223$

Monotone decreasing, with ratio $ð^K(4,7)/ð^K(2,3) = 0.6374$ matching Paper 38 [43] single-factor bit-identical.

5 The Mondino–Sämann bridge: Wick-rotation functor

5.1 The F2 mismatch and three candidate resolutions

The structural mismatch between the Krein metametric $Đ^K$ (4-point relaxed forward, codomain $[0,\infty]$) and the Mondino–Sämann time separation ℓ (3-point reverse, codomain $\{-\infty\} \cup [0,\infty]$) is the **F2 mismatch**. We test three candidate resolutions.

5.1.1 R1: Negate time

Candidate. Define $\ell^L_{R1}(\hat\omega,\hat\omega') := -Đ^K(\mathbb{X}^K_{\hat\omega}, \mathbb{X}^K_{\hat\omega'})$.

Verdict: NEGATIVE. R1 fails for two independent reasons: (i) Codomain mismatch: $Đ^K \in [0,\infty]$ gives $\ell^L_{R1} \in [-\infty, 0]$, while Mondino–Sämann requires $\{-\infty\} \cup [0,\infty]$. The MS reverse triangle encodes the Lorentzian convention that "longer causal curves accumulate more proper time" and the negation gives the opposite convention. (ii) Triangle structural mismatch: negating both sides of the 4-point relaxed forward inequality does NOT collapse to the 3-point reverse triangle, and the coefficient factors $(1+Đ^K)^2$ have no MS-side analog.

5.1.2 R2: Functorial composite of Wick-rotation + modular-flow

Candidate. A bridge functor $F_a \circ F_c$ composed of: (a) Wick rotation F_a giving the right codomain shape (3-point reverse triangle on boost orbits via analytic continuation), and (c) modular flow F_c giving the right basepoint (BW vacuum) and the right cover scales ($\beta_k = 2\pi/\kappa_g$).

Verdict: PARTIAL. (a) gives the right structural shape but does not determine basepoint or cover scales. (c) gives basepoint and scales but inherits the F2 triangle-direction mismatch (modular flow is additive, not super-additive). The composite $F_a \circ F_c$ has all needed ingredients but the F2 mismatch is not resolved *by R2 alone*; the resolution requires R3.

5.1.3 R3: Connes–Rovelli thermal time as unifying framework

Candidate. Identify the Krein metametric $Ð^K$ as measuring *thermal time* of the wedge KMS state and the Mondino–Sämann ℓ as measuring *geometric proper time* of the same wedge KMS state. The Wick rotation between them, via equation (2), effects the analytic continuation $t_{\text{thermal}} = -it_{\text{geometric}}/\kappa_g$ mapping additive thermal-time structure to super-additive geometric-time structure.

Verdict: POSITIVE. R3 supplies the unifying interpretation that resolves F2. Modular flow $\sigma^{\omega}_{t_1} \circ \sigma^{\omega}_{t_2} = \sigma^{\omega}_{t_1+t_2}$ is *additive* in thermal time (forward-triangle), while geometric proper time on a Lorentzian manifold is *super-additive* on causal curves (the "twin paradox"). The two triangle directions are not "mismatched": they encode the same wedge KMS state under two distinct projection mechanisms (operator-algebraic modular flow vs. synthetic causal geometry), connected by the Wick rotation.

5.2 Definition of the Wick-rotation functor

Definition 5.1 (The category $\mathbf{KreinMetaMet}_{\text{pp}}$)**.** Let $\mathbf{KreinMetaMet}_{\text{pp}}$ be the category whose:

- Objects are Krein PPQMS $\mathbb{X}^K = (\mathcal{A}^K, L^K, \mathcal{M}^L, \omega^L_W)$ per Definition 3.13;
- Morphisms are topographic Krein M-isometries $\pi^K : \mathbb{X}^K_1 \to \mathbb{X}^K_2$, i.e. proper *-epimorphisms preserving topography and pin state.

Definition 5.2 (The category $\mathbf{LorPLG}_{\text{cov}}$)**.** Let $\mathbf{LorPLG}_{\text{cov}}$ be the category whose:

- Objects are covered LPLS $(X, \ell, o, \mathcal{U})$ per Definition 2.9;
- Morphisms are ℓ-preserving maps preserving basepoint o and cover $\mathcal{U}$.

Definition 5.3 (The Wick-rotation functor)**.** The *Wick-rotation functor* $\mathbf{W} : \mathbf{KreinMetaMet}_{\text{pp}} \to \mathbf{LorPLG}_{\text{cov}}$ is defined on objects by

$$\mathbf{W}(\mathcal{A}^K, L^K, \mathcal{M}^L, \omega^L_W) := (\hat{\mathcal{M}}^L, \ell^L, \hat{\omega}^L_W, \hat{\mathcal{U}}^L),$$

where:

1. $\hat{\mathcal{M}}^L := \operatorname{Spec}(\mathcal{M}^L)$ is the Gelfand spectrum of the M-diagonal Abelian topography;
2. $\hat{\omega}^L_W$ is the character of $\mathcal{M}^L$ induced by the BW vacuum;
3. $\hat{\mathcal{U}}^L = (\hat{U}_k)_{k\in\mathbb{N}}$ with $\hat{U}_k := \operatorname{Spec}(\mathcal{M}^L|_{(n_{\max}(k), N_t(k), T(k))})$ along admissible scaling per Paper 47 [51];
4. The time separation ℓ^L on $\hat{\mathcal{M}}^L$ is

$$\ell^L(\hat{\omega}, \hat{\omega}') := \begin{cases} \kappa_g \cdot \tau^{\omega^L_W}_{\text{mod}}(\hat{\omega}, \hat{\omega}') & \text{if } \hat{\omega} \preceq \hat{\omega}' \text{ along modular flow,} \\ -\infty & \text{otherwise.} \end{cases}$$

On morphisms, $\mathbf{W}(\pi^K) := \hat{\pi}^K$ is the dual Gelfand spectrum map (contravariant by Gelfand duality: $\mathbf{KreinMetaMet}_{\text{pp}} \to \mathbf{LorPLG}_{\text{cov}}$ on objects sends $\mathbb{X}^K$ forward but on morphisms the arrow direction reverses).

5.3 The Krein–Mondino–Sämann Bridge Theorem

Theorem 5.4 (Krein–Mondino–Sämann Bridge). *The Wick-rotation functor* $\mathbf{W}$ *of Definition 5.3 is well-defined, with the following four properties:*

(B1) Structural correspondence. $\mathbf{W}$ *sends the Krein PPQMS substrate to the Mondino–Sämann covered LPLS structure row-by-row as in Table 1.*

(B2) Reverse triangle inequality. *For every* $\mathbb{X}^K$ *and every* $\hat{\omega}_x, \hat{\omega}_y, \hat{\omega}_z \in \mathbf{W}(\mathbb{X}^K)$,

$$\ell^L(\hat{\omega}_x, \hat{\omega}_y) + \ell^L(\hat{\omega}_y, \hat{\omega}_z) \leq \ell^L(\hat{\omega}_x, \hat{\omega}_z).$$

(B3) Pre-compactness inheritance. *The Mondino–Sämann pre-compactness conditions (Thm 6.2 of MS) hold on* $\mathbf{W}(\mathbb{X}^K_n)$ *for any sequence of truncated Krein PPQMS* $(\mathbb{X}^K_n)$ *at admissible-scaling cutoffs.*

(B4) Convergence transport. *If* $Đ^K(\mathbb{X}^K_n, \mathbb{X}^K_\infty) \to 0$, *then* $\mathbf{W}(\mathbb{X}^K_n) \to \mathbf{W}(\mathbb{X}^K_\infty)$ *in the Mondino–Sämann pLGH sense.*

5.4 Structural correspondence (B1)

The structural correspondence holds row-by-row per Table 1:

Table 1: The structural correspondence (B1) between Mondino–Sämann covered LPLS and Krein PPQMS structures, exhibited by the Wick-rotation functor $\mathbf{W}$.

#	Mondino–Sämann (synthetic)	Krein-algebraic-side analog
1	Underlying set X	$\hat{\mathcal{M}}^L = \mathrm{Spec}(\mathcal{M}^L)$
2	Topology finer than chronological	weak-$*$ topology on $\hat{\mathcal{M}}^L$
3	Time separation ℓ	Wick-rotated $\ell^L = \kappa_g \cdot \tau_{\mathrm{mod}}$
4	Reverse triangle (Eq. (1))	on orbit: equality; off orbit: trivial $-\infty$
5	Timelike $\ll$	modular precedence with $t > 0$
6	Causal $\leq$	modular precedence with $t \geq 0$
7	Causal diamond	$\{\hat{\omega}_x \circ \sigma_s : 0 \leq s \leq t\}$
8	Basepoint event o	BW vacuum $\hat{\omega}^L_W$
9	Cover $\mathcal{U}$	truncated topography spectra $(\hat{U}_k)$
10	Cover scales β_k	BW canonical period $\beta_k = 2\pi/\kappa_g$
11	Slab bound	uniform 2π via modular periodicity
12	ε-net	modular causal diamond ε-net at scale $\gamma^{\mathrm{joint}}(k)$
13	LGH convergence	truncation-projector correspondence
14	pLGH convergence	per-cover LGH convergence
15	Pre-compactness	cardinality bound from propagation $= 2$

5.5 Reverse triangle inequality (B2): Decomposition O on the M-diagonal topography

The substantive content of (B2) is the resolution of F2 via R3 plus the structural simplification that off-orbit cases are trivial at the K^+-weak-form bridge.

Lemma 5.5 (Modular flow acts trivially on the topography). *For every topographic generator* $a = M^{\mathrm{spat}}_{N,L,0} \otimes I_{N_t} \in \mathcal{M}^L$,

$$\sigma_t^{\omega^L_W}(a) = a \quad \forall t \in \mathbb{R}.$$

Consequently, the dual action on $\hat{\mathcal{M}}^L$ *is trivial:* $\hat{\sigma}_t^{\omega^L_W}(\hat{\omega}) = \hat{\omega}$ *for all* $\hat{\omega}$.

Proof. $[K_\alpha^W, M^{\text{spat}}_{N,L,0}] = 0$ because K_α^W is M-diagonal and the topographic generator has $M = 0$, both diagonal in the same (N, L, M)-block decomposition. Hence $\sigma_t^{\omega_W^L}(a) = e^{-itK_\alpha^W} a e^{+itK_\alpha^W} = a$, and the dual Gelfand action is trivial. $\square$

Lemma 5.6 (Decomposition O for triples of characters). *For $\hat{\omega}_x, \hat{\omega}_y, \hat{\omega}_z \in \hat{\mathcal{M}}^L$ at finite cutoff, indexed by topographic eigenvalue labels $(N_x, L_x), (N_y, L_y), (N_z, L_z)$ with $M = 0$, the off-orbit case decomposes into four sub-cases:*

(i) All three on the same modular-flow orbit: $(N_x, L_x) = (N_y, L_y) = (N_z, L_z)$. On-orbit case;

(ii) $\hat{\omega}_x, \hat{\omega}_z$ on the same orbit, $\hat{\omega}_y$ on a different orbit. Then $\ell^L(\hat{\omega}_x, \hat{\omega}_y) = -\infty$ (or $\ell^L(\hat{\omega}_y, \hat{\omega}_z) = -\infty$);

*(iii) $\hat{\omega}_x, \hat{\omega}_y$ on one orbit, $\hat{\omega}_y, \hat{\omega}_z$ on a different orbit, $\hat{\omega}_x, \hat{\omega}_z$ on yet a third. **Empty** (this would require $(N_x, L_x) = (N_y, L_y) \neq (N_z, L_z) = (N_x, L_x)$, a contradiction);*

(iv) All three on different orbits. At least one $\ell^L = -\infty$.

Proof sketch. The orbit-preservation of modular flow (Lemma 5.5) means modular flow on $\hat{\mathcal{M}}^L$ preserves the (N, L) labels. Hence two characters are causally related ($\hat{\omega} \preceq \hat{\omega}'$ along modular flow with finite τ_{mod}) only if they share the same (N, L) label. Cases (i)–(iv) are then a complete enumeration of label-tuple configurations, and (iii) is structurally empty. $\square$

Proof of Theorem 5.4 (B2). By Lemma 5.6, we analyze (B2) case by case.

Case (i), on-orbit (equality). If $\hat{\omega}_x \preceq \hat{\omega}_y \preceq \hat{\omega}_z$ along modular flow with $\hat{\omega}_y = \hat{\omega}_x \circ \sigma_{t_1}^{\omega_W^L}$ and $\hat{\omega}_z = \hat{\omega}_y \circ \sigma_{t_2}^{\omega_W^L}$, then $\hat{\omega}_z = \hat{\omega}_x \circ \sigma_{t_1+t_2}^{\omega_W^L}$ by additivity of modular flow. Hence $\ell^L(\hat{\omega}_x, \hat{\omega}_y) + \ell^L(\hat{\omega}_y, \hat{\omega}_z) = \kappa_g(t_1 + t_2) = \ell^L(\hat{\omega}_x, \hat{\omega}_z)$.

Cases (ii), (iv), trivial $-\infty$ inheritance. At least one of $\ell^L(\hat{\omega}_x, \hat{\omega}_y)$ or $\ell^L(\hat{\omega}_y, \hat{\omega}_z)$ is $-\infty$, so the LHS is $-\infty$ by MS convention, and $-\infty \leq \ell^L(\hat{\omega}_x, \hat{\omega}_z)$ trivially.

Case (iii), empty. No contribution.

Hence (B2) holds in all cases. $\square$

Remark 5.7 (The super-additivity that DOES exist lives in $\mathcal{O}^L$, not in $\mathcal{M}^L$). The substantive super-additivity property (off-axis Lorentz-boost composition accumulating more proper time than direct composition) exists on the wedge but lives in the *full* Krein operator system $\mathcal{O}^L$, not in the M-diagonal topography $\mathcal{M}^L$. The Paper 46 [50] Appendix B enlarged substrate (chirality- flipping generators M^{flip} with $\{J_L, M^{\text{flip}}\} = 0$) supports such off-axis composition; on this enlarged substrate, the bridge $\mathbf{W}$ would extend to a richer LPLS structure with strict super-additivity off-orbit. This is the open question Q1' (Section 8.1).

5.6 Pre-compactness inheritance (B3)

Proof of Theorem 5.4 (B3). The Mondino–Sämann pre-compactness Thm 6.2 has three conditions: (i) uniform slab bounds; (ii) uniform ε-net cardinality bounds; (iii) cumulative ε-net nesting.

(i) Uniform slab bound. The timelike diameter of $\hat{U}_k$ equals $\sup \tau_{\text{mod}}^{\omega_W^L}$ on the wedge. Under BW canonical $\kappa_g = 1$, modular flow is 2π-periodic (Paper 42 [46] § 5 integer spectrum of K_α^W), so $\tau_{\text{mod}} \leq 2\pi$ uniformly: $T_k = 2\pi$ for all k.

(ii) Uniform ε-net cardinality bound. At finite cutoff $(n_{\max}(k), N_t(k), T(k))$, $|\hat{U}_k| = O(n_{\max}(k)^4 \cdot N_t(k))$ (dimension of the topography), and the ε-net cardinality is bounded above by $|\hat{U}_k|$ itself and by the standard discrete-to-continuous refinement bound $N(k, \varepsilon) = \min(|\hat{U}_k|, \text{Vol}(\hat{U}_k)/\varepsilon^{d_k})$. Both bounds are uniform via Paper 44 [48] propagation number $= 2$.

(iii) Cumulative ε-net nesting. By the admissible-scaling sequence $(n_{\max}(k), N_t(k), T(k)) \to (\infty, \infty, T_\infty)$, the cover $\hat{U}_k \subseteq \hat{U}_{k+1}$ is nested by construction, and ε-nets of $\hat{U}_k$ extend to ε-nets of $\hat{U}_{k+1}$ by adding finitely many points.

By Mondino–Sämann Thm 6.2, the bridge image inherits pre-compactness. □

5.7 Convergence transport (B4) via Plancherel-nested Berezin

Theorem 5.8 (Convergence transport)**.** *If $Ð^K(\mathbb{X}_n^K, \mathbb{X}_\infty^K) \to 0$ in the Krein hypertopology of Definition 4.8, then $\mathbf{W}(\mathbb{X}_n^K) \to \mathbf{W}(\mathbb{X}_\infty^K)$ in the Mondino–Sämann pLGH topology of Definition 2.10, with the strong (timelike forward density) sense.*

Proof sketch. By the Krein hypertopology coincidence (Theorem 4.5) plus the bit-exact panel inheritance from Paper 45 [49], the Krein-side propinquity convergence implies that for each cover level k, the truncated topography spectra $\hat{U}_{k,n} \to \hat{U}_{k,\infty}$ in the LGH sense (per Definition 2.8). The Plancherel-nested Berezin/projection pair from Paper 45 [49] Sub-Sprint C/D supplies a Mondino–Sämann correspondence with vanishing distortion. Per-axiom verification:

Cardinality matching (MS Def 3.6(i)). At fixed admissible cutoff, the correspondence pair has $|S_n| = |S_\infty|$ on the M-diagonal topography (the Berezin map preserves the (N, L) shell structure).

Correspondence with vanishing distortion (MS Def 3.6(ii)). The Berezin/projection pair has reach $\leq \gamma^{\text{joint}}$ + height $\leq \gamma^{\text{joint}} \cdot C_3^{\text{joint}}$, vanishing as $n \to \infty$.

Extension property (MS Def 3.6(iii)). The Plancherel-nested structure of Berezin maps IS the MS extension property: Berezin maps at finer cutoff extend Berezin maps at coarser cutoff because the Plancherel weights nest by construction (Paper 45 [49] Sub-Sprint C Lemma 5.1).

Forward density (MS Def 3.6(iv)). The modular-flow orbit discretization in the chronological topology provides the timelike forward density: for any $\hat{\omega}' \succ \hat{\omega}$ on a modular orbit, the truncation projects $\hat{\omega}'$ to a sequence $\hat{\omega}'_n$ converging to $\hat{\omega}'$ in the chronological topology with $\hat{\omega} \prec \hat{\omega}'_n$ for all sufficiently large n.

Per-cover LGH convergence at each k implies pLGH convergence per Definition 2.10. □

5.8 Riemannian-limit recovery cross-check

Corollary 5.9. *At $N_t = 1$, the Krein-side propinquity bound $\Lambda^L_{n_{\max},1,T} = \Lambda^{\text{Paper 38}}_{n_{\max}}$ bit-exact (Paper 45 Sub-Sprint D Lemma 5.1), so the bridge image $\mathbf{W}(\mathbb{X}^K|_{N_t=1})$ is the synthetic-side counterpart of Paper 38's SU(2) Riemannian propinquity: a trivial LPLS with no Lorentzian content (modular flow on the Riemannian limit collapses to identity, and $\ell^L|_{N_t=1} = 0$ along the Gelfand spectrum).*

5.9 Honest scope of the bridge theorem

The bridge functor $\mathbf{W}$ is constructed at the K^+-weak-form level on the natural chirality-doubled scalar-multiplier substrate of Papers 44/45. The strong-form extension of $\mathbf{W}$ to the Paper 46 [50] Appendix B enlarged substrate (chirality-flipping generators) is **not** constructed in this paper; it is the open question Q1' (Section 8.1).

The bridge is **functorial, not isometric**: $Ð^K$ and ℓ^L measure different physical quantities (thermal vs. geometric time) of the same wedge KMS state. The two triangle directions are not "mismatched" — they encode different projections of the same physical object, connected by the Wick rotation under R3 (Connes–Rovelli thermal time).

6 Application to the GeoVac hemispheric wedge

6.1 The GeoVac wedge as a Krein PPQMS

Lemma 6.1 (The wedge as Krein PPQMS). *The Paper 43 [47] hemispheric wedge $W_L = P_W^{\text{spat}} \otimes P_{t\geq 0}$ at every finite cutoff $(n_{\max}, N_t, T)$ defines a Krein PPQMS*

$$\mathbb{X}^K_{\text{wedge}} := (\mathcal{A}^L|_{W_L}, L^K|_{W_L}, \mathcal{M}^L|_{W_L}, \omega^L_W),$$

with all five Krein PPQMS axioms of Definition 3.13 verified by direct construction.

Proof sketch. (A1) $\mathcal{A}^L|_{W_L}$ is the wedge-restricted Lorentzian operator system (Paper 44 [48] § 2). (A2) $L^K|_{W_L}$ is the K$^+$-restricted Krein-Leibniz Lipschitz seminorm of Definition 3.1. (A3) $\mathcal{M}^L|_{W_L}$ is the wedge-restricted M-diagonal Abelian topography of Lemma 3.12. (A4) $\omega^L_W = e^{-K^W_\alpha}/Z$ is the wedge KMS state (Paper 43 [47] § 4.2), diagonal in the M-eigenbasis hence a character of $\mathcal{M}^L|_{W_L}$. (A5) The Krein M-tunnel structure with $\chi^K \to 0$ is the wedge restriction of Lemma 4.2. □

6.2 The seven newly accessible theorems

We apply the bridge functor $\mathbf{W}$ to the wedge PPQMS:

$$\mathbf{W}(\mathbb{X}^K_{\text{wedge}}) = (\hat{\mathcal{M}}^L|_{W_L}, \ell^L_{\text{wedge}}, \hat{\omega}^L_W, \hat{\mathcal{U}}^L_{\text{wedge}}),$$

where at panel cells the cardinality is $|\hat{\mathcal{M}}^L|_{W_L}| = \dim W_L = 16, 60, 160$ at $(n_{\max}, N_t) \in \{(2,3), (3,5), (4,7)\}$ respectively.

6.2.1 T1: The wedge as Mondino–Sämann covered LPLS

Theorem 6.2 (T1: GeoVac wedge as Mondino–Sämann pointed pre-length space). *For every finite cutoff $(n_{\max}, N_t, T)$ with $T = 2\pi$ canonical, the GeoVac hemispheric wedge W_L defines a Mondino–Sämann covered Lorentzian pre-length space*

$$(\hat{\mathcal{M}}^L|_{W_L}, \ell^L_{\text{wedge}}, \hat{\omega}^L_W, \hat{\mathcal{U}}^L_{\text{wedge}}) \in \mathbf{LorPLG}_{\text{cov}},$$

with uniform timelike diameter 2π across all cover levels.

Proof sketch. Direct application of Theorem 5.4 (B1) plus Lemma 6.1. The uniform timelike diameter follows from Paper 43 [47] Thm bw_alpha_lorentz: modular flow is 2π-periodic on the wedge. □

6.2.2 T2: Synthetic compactness of the wedge family

Theorem 6.3 (T2: GeoVac wedge synthetic compactness). *Any sequence $(\mathbb{X}^K_{\text{wedge}}(k_n))_{n\in\mathbb{N}}$ of GeoVac wedge Krein PPQMS at admissible-scaling cutoffs $(n_{\max}(k_n), N_t(k_n), T(k_n)) \to (\infty, \infty, T_\infty)$ has a subsequence whose bridge images $\mathbf{W}(\mathbb{X}^K_{\text{wedge}}(k_n))$ converge in the Mondino–Sämann pLGH sense to a covered LPLS limit.*

Proof sketch. By Theorem 5.4 (B3), the three pre-compactness conditions of MS Thm 6.2 hold for the wedge sequence. The MS Thm 6.2 yields the pLGH-convergent subsequence. □

6.2.3 T3: Synthetic bit-exact panel inheritance

Theorem 6.4 (T3: Synthetic pLGH-convergence rate panel for the GeoVac wedge)**.** *The Mondino–Sämann pLGH-convergence rate of the GeoVac wedge sequence at the Paper 45 [49] panel cells satisfies:*

$(n_{\max}, N_t)$	$d^{\text{wedge}}_{\text{pLGH}}$ *(Krein-side Λ^L)*
$(2,3)$	≤ 2.0746
$(3,5)$	≤ 1.6101
$(4,7)$	≤ 1.3223

bit-identical to the Paper 45 Krein-side Λ^L values. This is the ***first quantitative pLGH-convergence panel in the Mondino–Sämann literature derived from operator-algebraic input.***

Proof sketch. The bridge $\mathbf{W}$ sends the Krein-side propinquity bound Λ^L to the synthetic-side pLGH rate d_{pLGH}. The functor is not isometric (Đ^K and ℓ^L measure different physical quantities), but the rate parameter $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ is the same object on both sides, hence the numerical panel transports bit-identically. □

Remark 6.5 (The substantive new content)**.** All extant Mondino–Sämann-style examples (arXiv:2504.10380 v4 § 10) are constructed geometrically from continuous spacetimes: Chruściél–Grant approximations of globally hyperbolic spacetimes, sequences of warped products, etc. Theorem 6.4 is the first quantitative pLGH-convergence rate panel in the MS literature derived from *operator-algebraic* input via the bridge.

6.2.4 T4: Synthetic four-witness readings

Theorem 6.6 (T4: Synthetic four-witness readings)**.** *The Paper 42 [46] four-witness Wick-rotation theorem identifies six physical thermal-time witnesses (Bisognano–Wichmann canonical, Hartle–Hawking at $M = 1, 2$, Sewell at $M = 1$, Unruh at $a = 1, 2$) on the operator-algebraic side, all giving bit-identical modular operators on the wedge KMS state. Via $\mathbf{W}$, these six physical witnesses give six geometric-time readings of the same synthetic-Lorentzian wedge pre-length space, distinguished only by the rapidity-rate normalization κ_g:*

Witness	κ_g	*Synthetic time scale*
BW canonical	1	$\ell^L = \tau_{\text{mod}}$
Hartle–Hawking $M = 1$	1/4	$\ell^L = \tau_{\text{mod}}/4$
Hartle–Hawking $M = 2$	1/8	$\ell^L = \tau_{\text{mod}}/8$
Sewell $M = 1$	1/4	*same as HH $M = 1$*
Unruh $a = 1$	1	*same as BW canonical*
Unruh $a = 2$	2	$\ell^L = 2\tau_{\text{mod}}$

Witnesses with the same κ_g produce identical synthetic LPLS: the six physical instantiations on the operator side reduce to ***two synthetic-side equivalence classes****.*

Proof sketch. The bridge time separation $\ell^L = \kappa_g \cdot \tau_{\text{mod}}$ rescales linearly with κ_g. The bit-exact algebra-action collapse of the six witnesses on the Krein side (Paper 42 [46] § 8 Cor 8.1) translates to a bit-exact metric-rescaling identification on the synthetic side. □

6.2.5 T5: Synthetic Pythagorean orthogonality with $1/\pi^2$ M1 signature

Theorem 6.7 (T5: Synthetic-side Pythagorean orthogonality)**.** *Under the bridge* $\mathbf{W}$ *applied to the GeoVac hemispheric wedge at the Riemannian limit* $N_t = 1$*, the BW-aligned thermal-time generator* $H_{\text{local}} = K_\alpha^W/\beta$ *and the wedge-restricted Lorentzian Dirac* D_W^L *correspond to two structurally orthogonal foliation structures on the synthetic wedge pre-length space: a boost-orbit foliation* $\mathcal{F}_\alpha$ *(*H_{local} *generator,* Π_W*-even per Paper 43 [47] Lem parity_HS input (I1)) and a spinor-bundle foliation* $\mathcal{F}_D$ *(*D_W^L *generator,* Π_W*-odd per input (I2)).*

The Paper 43 [47] § 10.2 Pythagorean closed-form residual

$$\|H_{\text{local}} - D_W^L\|_F^2 = \frac{\kappa_g^2 \cdot S(n_{\max})}{4\pi^2} + D(n_{\max})$$

translates to a synthetic-side decomposition of the wedge's geometric data into a κ_g^2*-scaling Hopf-base-measure contribution (*$S(n_{\max})/(4\pi^2)$*) and a* κ_g*-independent spinor-bundle contribution (*$D(n_{\max})$*). The* $1/\pi^2 = \text{Vol}(S^1)^{-2}$ *master Mellin engine M1 signature (Paper 32 [42] § VIII* `rem:pythagorean_m1_closure`*) identifies as the inverse-squared measure of the bridge cover's slab-period.*

Proof sketch. The algebraic identity is Paper 43 [47] Thm pythagorean_formal. The synthetic-side reading is by application of $\mathbf{W}$: the Π_W-grading on the Krein side translates to the orthogonal foliation structure on the synthetic side via Gelfand duality, and the $1/\pi^2$ prefactor on the operator-algebraic Pythagorean residual IS the synthetic-side wedge's slab-period inverse-squared measure ($\beta = 2\pi$, $1/\beta^2 = 1/(4\pi^2)$). □

6.2.6 T6: G2 metric-level closure for the GeoVac wedge

Theorem 6.8 (T6: G2 metric-level closure for the GeoVac hemispheric wedge)**.** *For the GeoVac hemispheric wedge specifically, the de-compactification limit* $T \to \infty$ *closes at the* propinquity-equivalent level on the synthetic side*: the synthetic wedge images* $\mathbf{W}(\mathbb{X}^K_{\text{wedge}}(k))$ *at admissible-scaling cutoffs converge in the Mondino–Sämann pLGH topology to the synthetic non-compact wedge object* $\mathbf{W}(\mathbb{X}^K_{\text{wedge}}(\infty))$ *(where* $\mathbb{X}^K_{\text{wedge}}(\infty)$ *is the bridge image of the Paper 47 [51] non-compact carrier* $S^3 \times \mathbb{R}_t$ *wedge).*

Equivalently, $d_{\text{pLGH}}(\mathbf{W}(\mathbb{X}^K_{\text{wedge}}(k)), \mathbf{W}(\mathbb{X}^K_{\text{wedge}}(\infty))) \to 0$ *as* $T(k) \to \infty$*.*

This is a ***GeoVac-wedge-specific closure of*** G2 *(de-compactification limit, Paper 45 [49] § 1.4) at the* ***metric-equivalent level on the synthetic side*** *that the general non-compact Latrémolière problem does not have at the operator side.*

Proof sketch. By Paper 47 [51] § 6, the three Krein-side carriers (periodic compact, bounded Dirichlet, non-compact $\mathbb{R}_t$) all converge in norm-resolvent sense to the same physical Lorentzian Dirac as $T \to \infty$. Under the bridge $\mathbf{W}$: $\mathbf{W}(\mathbb{X}^K_{\text{per}}(T))$, $\mathbf{W}(\mathbb{X}^K_{\text{Dir}}(T))$, and $\mathbf{W}(\mathbb{X}^K_{\mathbb{R}})$ are well-defined Mondino–Sämann pre-length spaces (T1 applies). Norm-resolvent convergence on the Krein side plus the bridge functoriality plus the (B2)+(B4) bridge properties give pLGH convergence on the synthetic side.

The substantive content is that the synthetic-side bridge image carries the structural content of metric-level convergence even though the Krein-side propinquity Λ^L is not defined on non-compact carriers (the Paper 45 propinquity requires compact temporal radius). □

6.2.7 T6a: Synthetic three-carrier identification

Corollary 6.9 (T6a: Synthetic three-carrier identification)**.** *The Paper 47 [51] § 6 three-carrier identification on the operator-algebraic side translates via the bridge* $\mathbf{W}$ *to a synthetic-side three-carrier identification: the three synthetic wedge images* $(\mathbf{W}(\mathbb{X}^K_{\text{per}}), \mathbf{W}(\mathbb{X}^K_{\text{Dir}}), \mathbf{W}(\mathbb{X}^K_{\mathbb{R}}))$ *are isometric as Mondino–Sämann covered LPLS in the limit* $T \to \infty$*.*

Proof sketch. The Paper 47 isometric embeddings $\widetilde{J}_T^{\mathrm{per}}, \widetilde{J}_T^{\mathrm{Dir}} : \mathcal{K}_T^{\bullet} \hookrightarrow \mathcal{K}_{\mathbb{R}}$ commute with the M-diagonal topographies (which are time-independent), hence descend via Gelfand duality to maps on the synthetic side. The synthetic three-carrier identification is the dual of the Krein-side identification. □

6.3 Honest scope of the wedge application

The wedge application is unconditional in the sense that the bridge applies cleanly to every load-bearing GeoVac structure (Paper 42 four-witness, Paper 43 wedge + Pythagorean, Paper 45 panel, Paper 47 three-carrier). The headline content (T6 G2 metric-equivalent closure for the wedge) is established at theorem-grade rigor for the wedge specifically; the general non-compact Latrémolière problem remains open. The strong-form bridge to the enlarged substrate is Q1' (Section 8.1).

7 Synthetic-Lorentzian compactness theorems

7.1 Synthetic compactness of the GeoVac wedge family

Theorem 6.3 (T2) provides the synthetic compactness statement for the GeoVac wedge family. We restate it in the context of synthetic-Lorentzian compactness theory.

Theorem 7.1 (Synthetic compactness restated)**.** *The class $\mathfrak{X}_{\text{GeoVac wedge}}$ of bridge images of GeoVac wedge Krein PPQMS at admissible-scaling cutoffs is pre-compact in the Mondino–Sämann pLGH topology of Definition 2.10.*

This is a substantive new content for the Mondino–Sämann literature: the wedge family is the first operator-algebraically derived class to satisfy the Mondino–Sämann pre-compactness criteria.

7.2 The first quantitative pLGH panel from operator-algebraic input

By Theorem 6.4 (T3), the GeoVac wedge family furnishes the first quantitative pLGH-convergence panel derived from operator-algebraic input. The bit-exact numerical values $\{2.0746, 1.6101, 1.3223\}$ across $(n_{\max}, N_t) \in \{(2,3), (3,5), (4,7)\}$ provide a concrete benchmark for future Mondino–Sämann-style synthetic-Lorentzian numerical work.

7.3 G2 metric-equivalent closure for the GeoVac wedge

The deepest substantive new content of the paper is Theorem 6.8 (T6). We unpack the structural significance:

- **The non-compact Latrémolière propinquity is not in the published literature.** The original Latrémolière propinquity [22, 24] and its 2025 hypertopology extension [25] operate on compact (or proper, locally compact) quantum metric spaces. A non-compact extension to the non-compact carrier $S^3 \times \mathbb{R}_t$ remains open. This is the published-open problem G2 metric-level of Paper 45 [49] § 1.4.
- **The bridge supplies a workaround at the wedge level.** The Wick-rotation functor $\mathbf{W}$ converts the operator-algebraic non-compact propinquity question to a synthetic-Lorentzian pLGH convergence question (Mondino–Sämann pLGH on non-compact LPLS is well-defined). The synthetic-side closure exists; the bridge transports it back to the operator side.

- **The closure is wedge-specific.** The bridge applies to any Krein PPQMS, but the headline content (G2 closure) requires the 2π-periodicity of modular flow on the wedge and the bridge's ability to extend along the admissible-scaling sequence to the non-compact carrier. For more general (non-wedge) carriers, the 2π-periodicity does not hold and the closure does not extend.

7.4 Synthetic three-carrier identification

Corollary 6.9 (T6a) extends the Paper 47 [51] three-carrier identification from the operator side to the synthetic side. This resolves an apparent tension between Papers 43 and 45's different temporal-carrier choices (bounded Dirichlet vs. periodic compact) at the synthetic level: both, plus the non-compact carrier, are isometric as covered LPLS in the de-compactification limit.

7.5 Why this is genuinely new content for Mondino–Sämann pLGH theory

The Mondino–Sämann pLGH theory [30] develops the synthetic side independently of operator algebras. All extant examples in arXiv:2504.10380 v4 § 10 are constructed geometrically: Chruściél–Grant approximations of globally hyperbolic spacetimes, warped products, etc. The bridge construction of this paper introduces a categorically new source of MS-style examples: operator-algebraic truncations transported to the synthetic side via the Wick-rotation functor. The substantive headlines (T3 quantitative panel, T6 G2 closure) are genuinely new content for the Mondino–Sämann literature.

7.6 Related synthetic-Lorentzian convergence frameworks

We position the bridge construction vs. adjacent synthetic-Lorentzian convergence frameworks. The Sormani–Vega null distance [36] and Sakovich–Sormani timed Gromov–Hausdorff [34] operate via null-distance metric-GH rather than via Mondino–Sämann's ε-net-of-causal-diamonds. Minguzzi–Suhr bounded Lorentzian metric spaces [28] extend a complementary distinction metric. Allen–Burtscher metric GH [1] and Müller's Cauchy slabs [31] provide alternative frameworks. The bridge of this paper targets specifically the Mondino–Sämann pLGH framework on covered LPLS; a parallel bridge to null-distance or timed-Hausdorff frameworks would be a separate construction.

8 Open questions

8.1 Q1': Strong-form bridge on the enlarged substrate

The most natural follow-on to the K^+-weak-form bridge of this paper is the strong-form bridge extending $\mathbf{W}$ to the Paper 46 [50] Appendix B enlarged substrate (chirality-flipping generators M^{flip} with $\{J_L, M^{\mathrm{flip}}\} = 0$).

> **Open question Q1' (strong-form Krein–MS bridge).** *Does the Wick-rotation functor* $\mathbf{W}$ *extend to the enlarged substrate of Paper 46 [50] Appendix B, and if so, does the off-orbit super-additivity close via direct transport of the Paper 42 [46] four-witness theorem to MS time separation?*

The substantive question that A.4'-A surfaced

On the K^+-weak-form bridge of this paper, off-orbit super-additivity is structurally vacuous via Lemma 5.6 (Case (iii) is empty under the M-diagonal topography). On the strong-form bridge, the enlarged topography makes Case (iii) non-empty: there are triples $(\hat{\omega}_x, \hat{\omega}_y, \hat{\omega}_z)$ such that all three pairs are on different orbits with non-trivial chirality-flipping content.

In this regime, the operator-algebraic super-additivity from Paper 42's four-witness theorem becomes load-bearing: the on-orbit equality $\ell^L(x,y) + \ell^L(y,z) = \ell^L(x,z)$ generalizes to a strict super-additivity $\ell^L(x,y) + \ell^L(y,z) \leq \ell^L(x,z)$ via the analog of the special-relativistic twin paradox at the operator-algebraic level.

Q1' three-step decomposition

(Q1'.A) **Topography enlargement.** Extend the topography $\mathcal{M}^L$ to include M^{flip} generators; verify the topography axioms (Definition 3.11) hold for the enlarged structure. Substantive risk: the enlargement may break Abelianness (chirality-flipping generators do not commute with each other in general), in which case the Gelfand-spectrum construction does not apply directly.

(Q1'.B) **Gelfand spectrum.** Verify $\hat{\mathcal{M}}^L_{\mathrm{enlarged}}$ is well-defined. If Abelianness breaks: generalize to non-commutative Mondino–Sämann pre-length spaces (a substantial NCG-research target not currently in the literature).

(Q1'.C) **Off-orbit super-additivity.** Prove $\ell^L(x,y) + \ell^L(y,z) \leq \ell^L(x,z)$ at strict-inequality level via Paper 42 four-witness theorem transport to off-axis modular flow.

Q1' effort estimate and recommendation

Estimated effort: 3–6 months at sprint cadence. If Q1'.B forces the non-commutative MS extension route, the timeline expands further (the non-commutative MS pre-length space concept does not exist in the published literature; constructing it would itself be a 6–12 month NCG-mathematics target). Q1' is named here as the principal multi-month follow-on; it is the natural target for a Sprint L3e-Q1' or for an independent NCG-research program.

8.2 G2 metric-level closure on the general (non-wedge) carrier

Theorem 6.8 (T6) closes G2 metric-level for the GeoVac wedge specifically via the bridge image on the synthetic side. The general non-compact Latrémolière propinquity problem (Paper 45 [49] § 1.4 G2-metric on arbitrary non-wedge carriers) remains the multi-month NCG-mathematics frontier. Paper 47 [51] establishes the analog closure at the norm-resolvent / spectral level on the general carrier; the metric-level analog requires either:

- A non-compact Latrémolière propinquity extension (a multi-month NCG-math target); or
- A bridge to a non-wedge synthetic-Lorentzian framework that admits metric-level pLGH convergence on the general carrier (structurally distinct from the wedge bridge here).

8.3 G3 cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}}(S^5)$

The cross-manifold extension of the Lorentzian Krein spectral triple program is blocked at the NCG-framework level by the Paper 24 [40] § V four-layer Coulomb/HO asymmetry, including the (layer iv) modular-Hamiltonian structure of the wedge KMS state surfaced by Sprint L2-F.1. This obstruction is orthogonal to the present bridge: G3 is a structural extension target for the NCG framework itself, not for the bridge to Mondino–Sämann.

8.4 G4 inner-factor calibration data

The W3 spectral-zeta candidate for inner-factor calibration data was falsified (CLAUDE.md Sprint W3, 2026-05-08: CKM Wolfenstein candidate; three follow-up tracks ruled out PMNS, lepton masses, and the mechanically-generated basis). No concrete proposals for inner-factor calibration data remain in the literature. G4 is orthogonal to the present bridge.

8.5 Q2': Non-commutative Mondino–Sämann pre-length spaces

If Q1'.B (Section 8.1) forces breaking the Abelianness of the enlarged topography, the bridge construction would naturally generalize to non-commutative MS pre-length spaces. This is a substantial NCG-research target not currently in the literature.

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